

## Tutorial No : 3

**Tutorial Title:** Write a program in C++ to sort the numbers in an array using separate functions for read, display, sort and swap. The objective of this assignment is to learn the concepts of input, output, functions, call by reference in C++.

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### Program:

```
#include <iostream>
using namespace std;

void accept(int &, int []);
void display(int, int []);
void bubbleSort(int, int []);
void swapElements(int &, int &);

int main() {
    int n, array[10];
    accept(n, array);
    bubbleSort(n, array);
    cout << "\nSorted Array:";
    display(n, array);
    return 0;
}

void accept(int &m, int arr1[]) {
    cout << "\nEnter size of array: ";
    cin >> m;
    cout << "\nEnter array elements:\n";
    for (int i = 0; i < m; i++)
        cin >> arr1[i];
}

void display(int m, int arr1[]) {
    for (int k = 0; k < m; k++)
        cout << arr1[k] << " ";
}

void swapElements(int &a, int &b) {
    int temp = a;
```

```
    a = b;
    b = temp;
}

void bubbleSort(int n, int arr[]) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - 1 - i; j++) {
            if (arr[j] > arr[j + 1]) {
                swapElements(arr[j], arr[j + 1]);
            }
        }
    }
}
```

**Output:**

### Output

Enter size of array: 6

Enter array elements:

12 11 14 2 4 19

Sorted Array: 2 4 11 12 14 19

=== Code Execution Successful ===